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IPv6-Only and IPv6-Mostly Terminology ~~Definitions~~
draft-ietf-v6ops-ipv6-only-05

Commenté [MB1]: The document includes more than that. I wonder whether this can be generalized?

Abstract

This document defines the terminology regarding the usage of expressions such as "IPv6-Only" and "IPv6-Mostly"~~, in order~~ to avoid confusions when using them in IETF and other documents. In particular, ~~The~~ a goal of this document is ~~that~~ to provide a reference ~~to "IPv6-Only"~~that describes the actual "IPv6-Only" functionality being used in a given scope, not the installed protocol stack support.

Status of This Memo

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1. Introduction

~~Due~~ ~~Given to~~ the diverse and heterogenous systems that compose ~~nature of~~ the Internet and the ~~different~~ various types of users, distinct parts of a network, providers, flows, etc., there is not a single and easy way to categorically ~~say~~ express something such as "IPv6-Only".

~~The~~ One of the goals of this document is to depict this situation and ~~agree~~ define ~~in~~ a common language to be used ~~for~~ in the IETF and other documents~~, in order~~ to facilitate ~~ourselves and future readers,~~ conveying the correct understanding of IPv6 deployment related terms for authors and readers ~~what we are talking about~~.

For example, ~~The~~ the term "IPv6-Only" is ~~being~~ used ~~by~~ in many IETF documents~~,~~ without a clear definition of the scope or ~~terminology~~exact characterization, ~~for example~~such as [RFC6877], [RFC8585], and [RFC8683].

~~Note that a~~All the references in this document are ~~regarding~~ about the actual usage of IPv4/IPv6, not the support of those protocols by networks, nodes, and their components. For example, ~~a device or~~an access network may support both IPv4 and IPv6, however ~~actually~~ that network ~~is~~ might be only "natively" forwarding IPv6 packets~~,~~ if ~~because~~ the links used for ~~that communication~~forwarding ~~is~~ are only natively

Commenté [MB2]: To cover other scopes (interface, container, etc.)

configured for IPv6. IPv4 may be used as well~~,~~ but ~~it~~ IPv4 packets ~~is~~ are ~~being~~
encapsulated (or translated to) ~~by means of~~over IPv6. ~~So, f~~From this perspective, ~~this a device~~node connected to such a network ~~–~~is attached to ~~an~~ IPv6-Only link(s).

~~As such, a~~A network service is considered IPv6-Only if it forwards plain IPv6 packets, not IPv4 packets, even if IPv4 is still supported and enabled but not
configured neither used in the nodes participating in the service.

This document focuses on key terms that are used to denote main IPv4/IPv6 deployment models.

2. Precisely D~~d~~efined ~~scope~~ Applicability Scope ~~is~~ Is a ~~must~~Must

The transition from IPv4 to IPv6 is ~~not something that can~~ cannot be done,
in the large majority of the cases, overnight and in a single step in a ~~complete~~ network. Consequently, in general, it is not possible to ~~talk~~ reason about a whole network having a "single and uniform" status regarding the IPv6/IPv4 support, at least not in the early IPv6 deployment
stages of ~~an operator~~a network.

Even if possible, it is not frequent to deploy new end-site IPv6 networks which have no IPv4 connectivity at all, because at the current phase of the universal goal of the IPv6 deployment, almost every end-site network still needs to provide some kind of "access" to IPv4(-only) sites and services. It is not feasible for most of the operators to ~~tell~~ communicate to their customers "I can provide you IPv6
service, but you will not be able to access all Internet contents and apps, because some of them still don't support IPv6, so you will miss every content that it is IPv4-only". Of course, this will change over ~~the~~ time, and there will be less and less dependency on IPv4-only end-sites/services.

Some networks may have IPv6-Only support for specific purposes or services. For example, a DOCSIS provider may ~~have~~ decide~~d~~ that is worth the effort to get rid of IPv4 for the management network of the cable~~-~~ modems. ~~Or~~ Also, a network that provides connectivity only to IoT
devices, may be IPv6-Only; this is very common in Matter/Thread stub-networks.

However, the ~~"end-networks"~~stub networks, in general, need to continue supporting
IPv4, as there are many devices ~~or~~ and applications~~apps,~~ in both corporate and end-
user networks (smartTV, IP cameras, etc.), ~~which~~ that are IPv4-only and it
is not always feasible to update, upgrade, or replace them. Also, if customer
devices in a Local Area Network (LAN) are IPv4-only, they will not be able to access
IPv6-Only services~~, so this means that~~. Overall, IPv6-Only services can't be

Commenté [MB3]: Devices may be multi-attached

Commenté [MB4]: I don't parse this sentence.

Commenté [MB5]: Any reason why the convention used for IPv6-Only is not followed for IPv4-only?

Commenté [MB6]: Why do we need to have this?

Commenté [MB7]: Please add a reference.

deployed unless it is done in such way that some transition mechanism solves that problem as well (~~example~~ e.g., an IPv6-Only Data ~~Centre~~Center, requires SIIT-DC [RFC7755]).

In IPv6-Only access networks, IPv4 support may be provided by mechanisms that allow "IPv4-as-a-Service" (IPv4aaS), for example ~~by~~, by means of encapsulation ~~and/~~or translation on top of IPv6~~)~~.

In should be noticed ~~that~~ also that server segments are transitioning to IPv6-Only, in enterprise networks, both for servicing internal and external clients. In this case, to support IPv4-Only clients, multiple tools are available, such as stateless NAT64 with EAM [RFC7757]~~({RFC7757})~~ or SIIT-DC [RFC7755]~~({RFC7755})~~.

Consequently, considering the context described above, in order to be precise and avoid confusing readers by omitting necessary details, terminology such as "IPv6-Only" cannot be used in a generic way. Instead, an explicit indication of scope (e.g., what part of the network or which layer of the network stack is being referred to), must be provided. Therefore, it is necessary to define and describe the specific scope of applicability. The word "scope" is used in this precise sense throughout this document.

3. IPv4/IPv6 Native

"IPv4 Native" or "IPv6 Native" means that IP packets run directly over a layer 2 (logical link layer) interface, for example, IEEE 802.3 (Ethernet) link layer, without anything at layer 3 being encapsulated within an IP packet of another IP ~~protocol~~shim.

4. IPv4-Only

"IPv4-Only" in a given scope~~,~~ means that only IPv4 is native in that scope. IPv6 is neither configured or managed in that scope, even if it may be transported (or encapsulated) ~~on top of~~over IPv4.

5. IPv6-Only

"IPv6-Only" in a given scope~~,~~ means that only IPv6 is native in that scope. IPv4 is neither configured or managed in that scope, even if it may be transported (or encapsulated) ~~on top of~~over IPv6.

6. Dual-Stack

"Dual-Stack" in a given scope~~,~~ means that both~~,~~ IPv4 and IPv6 are native in that scope.

7. IPv4-as-a-Service~~IPv4 as a Service~~ (IPv4aaS)

In the case of IPv6-Only scopes, it is common that, to ensure that communications with legacy IPv4-Only scopes are possible, IPv4 packets may be transported ~~on top of~~over IPv6, typically by means of encapsulation or

Commenté [MB8]: Please expand

Commenté [MB9]: Please expand

Commenté [MB10]: There is no usage example of these terms in the applicability section.

translation. This can be made explicit by adding ~~"IPv4aaS" (IPv4 IPv4-~~as~~-~~
a Service (IPv4aaS). For example, IPv6-Only with IPv4aaS access network.

8. IPv6-Mostly (with/without DNS64)

"IPv6-Mostly" is similar to Dual-Stack, with two additional key elements that allow a host to act as IPv6-Only ~~({[I-D.ietf-v6ops-6mops]})~~: a stateful NAT64 ~~(~~[RFC6146]~~)~~ and DHCPv4 infrastructure operating Option 108 [RFC8925]~~({[RFC8925]})~~. Optionally, there may be also a DNS64 function ~~(~~[RFC6147]~~)~~. ~~This way~~In doing so, ~~in~~ a Dual-Stack network ~~scope, it~~ can support a mix of IPv4-~~only~~Only, Dual-Stack, ~~or~~ and IPv6-Only ~~clients~~endpoints, depending on the ~~client~~ endpoint's capabilities or configuration. ~~It~~IPv6-Mostly can be seen as ~~a way~~solution to provide IPv4 support on demand. Refer to [I-D.ietf-v6ops-6mops] for more architectural details.

Notes:

* There is no default interpretation regarding the DNS64 availability, so it is necessary to state "with" or "without DNS64" to explicitly qualify the case.

* When referring to a host, ~~it should be noted that~~ it should also have an embedded CLAT (464XLAT)~~,~~ [RFC6877]~~)~~ function.

* ~~The s~~Stateful NAT64 prefix~~es~~ should be learned by means ~~of~~ some mechanism, such as PREF64 Router Advertisement (RA) option [RFC8781], Heuristic Discovery [RFC7050]~~/~~[RFC8880] or Port Control Protocol (PCP) [RFC7225].

9. IPv6-Only-Strict

"IPv6-Only-Strict" in a given scope~~,~~ means that only IPv6 is native in that scope and IPv4 is neither configured or managed, but also not transported (neither encapsulated nor translated) ~~on top of~~over IPv6. In other words, it means that communication with other endpoints is only possible using IPv6.

10. Additional Scope Qualification

The scope can be further qualified providing additional information. A couple of examples are provided in the following subsections~~sub-sections~~.

10.1. ~~Control/Data-~~ Data, Control, and Management Plane Scope

In some cases, it may be relevant to indicate if referring to the data plane~~data-plane~~, control plane, management plane, ~~control-plane~~ or a combination thereof~~both~~.

10.2. API Scope

Commenté [MB11]: Maybe cite some examples, DS-Lite, MAP-E

Commenté [MB12]: Please update to use the bis

Commenté [MB13]: The definition may not stand if other means than DHCP option is used.

Maybe generalize and cite 108 as an example.

Commenté [MB14]: As this is the terminology used in I-D.ietf-v6ops-6mops

Commenté [MB15]: This is beyond terminology and all these details can be found in [I-D.ietf-v6ops-6mops] .

Do we need to have this listed here?

Commenté [MB16]: As there might be many

Commenté [MB17]: This is beyond terminology and all these details can be found in [I-D.ietf-v6ops-6mops] .

Do we need to have these listed here?

Commenté [MB18]: For consistency with the DOCSIS example provided above.

APIs can also be IPv4-only, IPv6-~~only~~Only, Dual-Stack or Multi-Stack (also supporting non-IP protocols). At the time of writing, networking APIs in operation systems are typically Dual-Stack or Multi-Stack regardless of any usage of IPv6-Only in other scopes. As a result ~~of this~~, applications usually operate correctly in any situation.

A typical example of a Dual-Stack feature in API scope is the getaddrinfo() function ~~(~~[RFC3493]~~)~~.

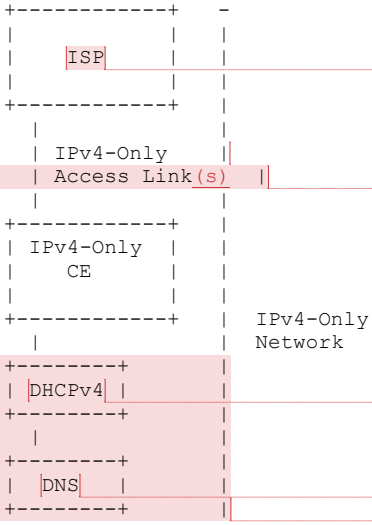
11. Example Diagrams

The ~~next~~ following example diagrams show only a few of the possibilities of different scope combinations. They have been chosen and ordered to follow the most natural approach, that in general, a "full" transition from IPv4 to IPv6 will follow. This doesn't mean that all these transition stages are ~~actually~~ needed in all the transition cases.

11.1. IPv4-Only Network

~~The following diagram~~Figure 1 shows the simplest case, when all the scopes are IPv4-Only and IPv6 is not present.

In general, an IPv4-Only network will have, even if embedded in the Customer Edge (CE) ~~Router~~ router~~(CE)~~, a DHCP~~v4~~ function, as well as a DNS resolver or DNS~~-~~ proxy. They are represented as different boxes in the diagrams, so it applies not just to residential and SOHO networks but also to enterprise networks.



Commenté [MB19]: Unless if they specifically request a specific AF not exposed

Commenté [MB20]: Or, add a statement early in the doc that DHCPv4 refers to DHCP [RFC2131]

Commenté [MB21]: ISP Network?

Commenté [MB22]: As there might be many

Commenté [MB23]: DHCPv4 Server/Relay?

Commenté [MB24]: DNS Server?

Commenté [MB25]: This is misleading as these can be interpreted as this are connected/linked, same path, etc.

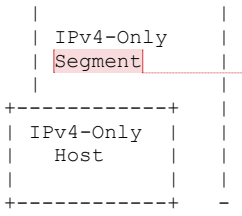


Figure 1: IPv4-Only diagram

11.2. Dual-Stack Network

~~The following diagram~~Figure 2 shows the simplest case, when all the scopes
are Dual-Stack.

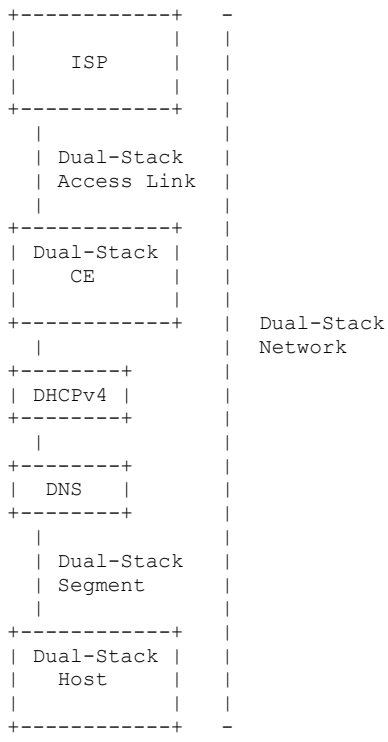


Figure 2: Dual-Stack diagram

11.3. IPv6-Only with IPv4aaS Network

~~The following diagram~~Figure 3 shows a very common case, ~~when~~ where the service
provider deploys IPv6-Only in the access link scope, instead of Dual-

Commenté [MB26]: I see what you mean but What's a segment? This is not used in the narrative text.

Commenté [MB27]: Add a note to explain how to interpret the links/Etc.

Commenté [MB28]: Same comments as those for Figure 1.

Commenté [MB29]: The doc uses service provider, operator, network provider, Internet Service Provider, etc. to refer to the same entity.

Please pick one and use it in the doc.

Stack, while the subscriber segments are still Dual-Stack. For example, ~~This~~ this is the 464XLAT case, which is also very frequent in mobile networks, where User Equipment (UE) and tethered devices remain Dual-Stack; here the UE will function as a router. The CE (or UE) will have a CLAT or equivalent function. The Internet Service Provider provides the stateful NAT64 (PLAT) function. ~~The end-hosts~~Endpoints may have a CLAT function, but they may not ~~turn it on~~enable it, because the CE already provides that functionality and DHCPv4 option 108 is not enabled/available.

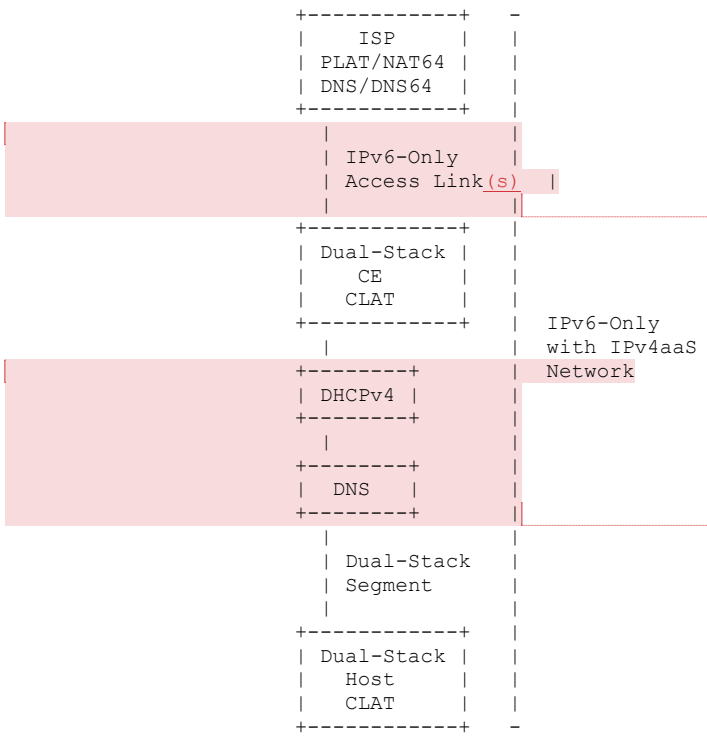


Figure 3: IPv6-Only with IPv4aaS diagram

11.4. IPv6-Mostly Segment with Dual-Stack Access Link

~~The following diagra~~Figure 4~~m~~ shows the case of IPv6-Mostly, which may be implemented in different ways. For example, when used in enterprises, the PLAT (and optionally the DNS64) may be implemented

a mis en forme : Surlignage

Commenté [MB30]: Please note in the «real» deployment this is not IPv6-only as there parts of the network that uses v4 encaps (GTP, for example).

Likewise, backbone may be using 6PE.

Note sure we need to call out all the subtelties but scoping is important

Commenté [MB31]: Idem as above.

in the enterprise network itself, in order to gain greater control of the configuration. It is important to note that, in this case, the DHCPv4 function needs to support ~~the~~ option 108. As a result, a Dual-Stack segment will be able to support both, hosts using IPv6-Only with a CLAT function (even if they are Dual-Stack), as well as IPv4-Only hosts or Dual-Stack hosts without the support of CLAT/ option 108. In residential and SOHO networks, typically the PLAT is located in the service provider network, as well as the DNS64. Using IPv6-Mostly in that case, restricts the communication between IPv4-Only hosts and those that are using IPv6-Only mode. This is resolved in enterprise networks by means of an internal PLAT with EAM [RFC7757], SIIT-DC [RFC7755], or other means. Residential, SOHO an enterprise networks could choose an IPv6-Only+IPv4aaS Access Link, so the case becomes a mix of them.

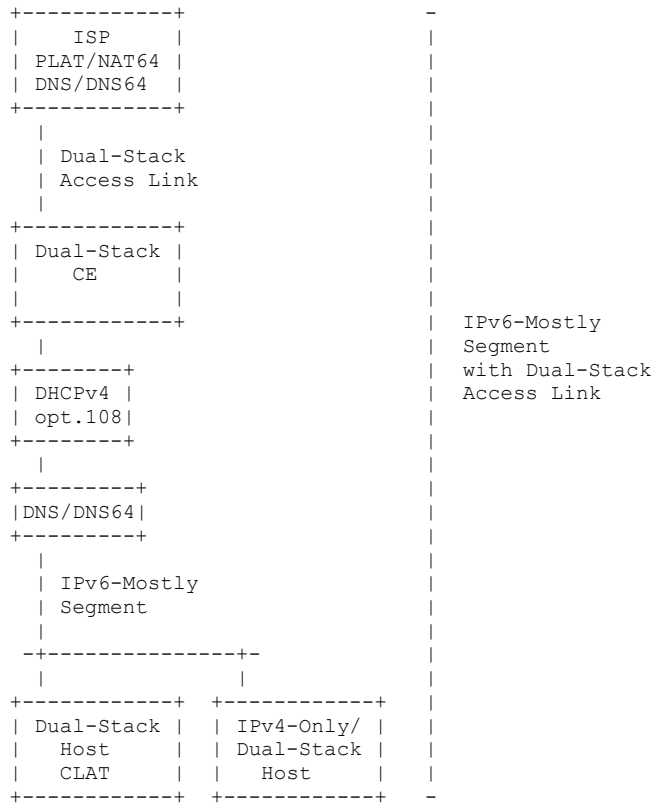


Figure 4: IPv6-Mostly diagram

Commenté [MB32]: Idem as previous figures

Overall, some explanation about how to «read» the figures would be helpful.

11.5. IPv6-Only Segment with Dual-Stack Access Link

~~The following diagram~~Figure 5 shows the case, more common in enterprise networks, when hosts in a segment are able to use IPv6-Only but they still need Dual-Stack access link because they still need to reach IPv4-Only in the Internet or other networks segments may need IPv4, for Example, if the enterprise network needs to expose some Dual-Stack services to Internet. In this case, DHCPv4 is not needed. Residential and SOHO networks, in general, will keep using an IPv6-Only Access Link.

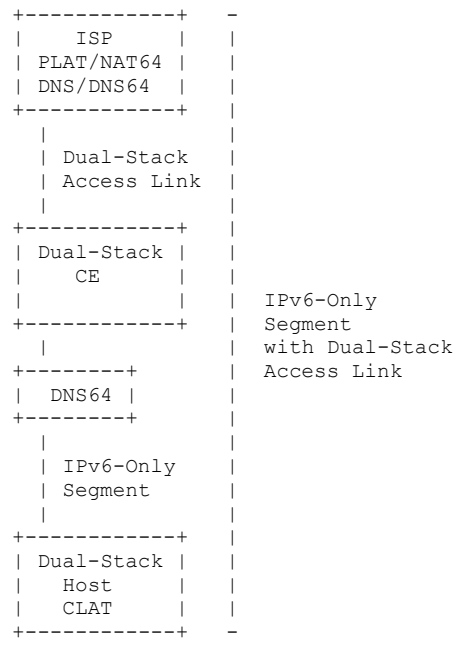


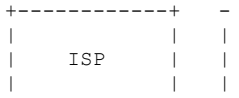
Figure 5: IPv6-Only diagram

Commenté [MB33]: Why?

Commenté [MB34]: Same comments as above.

11.6. IPv6-Only-Strict

~~The following diagram~~Figure 6 shows the case, when all the scopes of a network are using IPv6-Only and IPv4 communications are not possible. This would be the ideal "full transition" stage~~,~~ which, however, is not feasible in most of the cases, as maybe certain destinations in the Internet, that a network still needs to reach, will remain IPv4-Only for a long period of time.



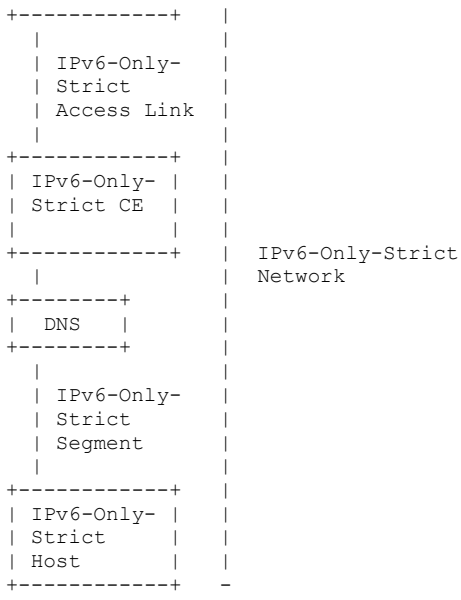


Figure 6: IPv6-Only-Strict diagram

12. Usage Examples and Practical ~~applicability~~Applicability

A typical example ~~will be~~ is a service provider network, which has different types of IPv4/IPv6 support in different parts of the network. Typically, it will be a "Dual-Stack core", "Dual-Stack upstream", "Dual-Stack BGP", "Dual-Stack router", but offers "IPv6-Only access". It is not common, but there ~~may~~ might be cases where it is
"IPv6-Only access ~~data-plane~~data plane" and "Dual-Stack access control~~-plane~~ plane"
or "IPv4-only access out-of-band management".

As of this writing, most end-user networks and hosts need to support IPv4, due to many global resources being only available over IPv4. Transition technologies may allow islands to be connected to the broader Internet over an IPv6-Only access networks acting as an underlay for legacy IPv4 traffic. An organization aiming to switch to an IPv6-Only end-user network will need to ensure that:

* ~~all~~ All host/routers are capable of IPv6-Only operation~~ and,~~.

* ~~all~~ All off-network resources are available over IPv6 (either as IPv6-Only or Dual-Stack).

For this reason, IPv6-Mostly seems an adequate ~~perfect~~ solution to easily~~,~~ ~~a~~
gradual move towards IPv6-Only (and IPv6-Only-Strict).

Commenté [MB35]: Same comments as previous figures

Commenté [MB36]: I wonder whether we can have some example to illustrate the scope usage at various levels:

Network, node, instance (containers, etc.), interface, link?

Commenté [MB37]: ???

Commenté [MB38]: What does this mean in BGP context? Is this about same session carrying both address families? Or two separate sessions?

The transport session does not matter much as both AFs can be exchanged over.

In the case of ~~data-centres~~data centers, "IPv6-Only compute nodes" may be provided with IPv4 external ~~IPv4~~ communication, using SIIT-DC. In this case, there is an "IPv6-Only ~~data-centre~~data center" when speaking about the internal LANs (anything behind ~~the~~ a Border Relay), but "Dual-Stack ~~data-centre~~data center upstreams".

In the case of a mobile network, the UEs (~~User Equipment,~~ e.g., mobile phones) have "IPv6-Only PDP Context", those UEs may offer "Dual-Stack Tethering" and "Dual-Stack to the UE applications"(by means of CLAT), and this is possible thanks to the support of a stateful NAT64 in the service provider network.~~; t~~ The stateful NAT64 is a "Dual-Stack service", which has an "IPv6-Only transport to the UEs".

Commenté [MB39]: Or PDU Session

In the case of an enterprise network, there may be a mix of VLANs or network segments offering Dual-Stack, IPv6-Only, and IPv6-Mostly. So in this case, it becomes clear by using "Dual-Stack VLAN x", "IPv6-Only VLAN y", and "IPv6-Mostly VLAN z".

IPv6-Only server, IPv6-Only ~~data-centre~~data center, and IPv6-Only cloud environments are entirely possible as of this writing, as long as:

* All host/routers are capable of IPv6-Only operation.

* All accessed resources (DNS resolvers, NTP servers, software update servers, network management services, and other external resources) are available over IPv6 (either IPv6-Only or Dual-Stack).

* All inbound communications are capable of IPv6, either due to all external endpoints supporting IPv6 or due to all legacy IPv4 traffic being relayed through a gateway (such as reverse proxy, SIIT-DC gateway, CDN, ~~etc~~etc.).

~~Data-centres~~Data centers and cloud environments may also support IPv6-Only subnets/interfaces and, at the same time, internal Dual-Stack in other subnets/interfaces, even if using non-globally reachable IPv4 addresses, such as those listed in Section 2.2.2 of [RFC6890], which may be also using NAT44.

13. Operational Considerations

This document provides definitions of various terms that are meant to unambiguously characterize specific deployment cases and would help tweak operational guidance for specific scopes.

~~13~~14. Security Considerations

This document does not have any specific security considerations.

14. IANA Considerations

This document does not have any IANA considerations.

15. Acknowledgements

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